

Problem Set 11
Due: May. 17, 2021

1. Section 4.5 Problem 41.

At a certain time, the number of people that enter an elevator is a Poisson random variable with parameter λ . The weight of each person is independent of every other person's weight, and is uniformly distributed between 100 and 200 lbs. Let X_i be the fraction of 100 by which the i th person exceeds 100 lbs, e.g., if the 7th person weighs 175 lbs., then $X_7 = 0.75$. Let Y be the sum of the X_i .

- (a) Find the transform associated with Y .
- (b) Use the transform to compute the expected value of Y .
- (c) Verify your answer to part (b) by using the law of iterated expectations.

Solution:

- (a) Let N be the number of people that enter the elevator. The corresponding transform is $M_N(s) = e^{\lambda(e^s - 1)}$. Let $M_X(s)$ be the common transform associated with the random variables X_i . Since X_i is uniformly distributed within $[0, 1]$, we have

$$M_X(s) = \frac{e^s - 1}{s}.$$

The transform $M_Y(s)$ is found by starting with the transform $M_N(s)$ and replacing each occurrence of e^s with $M_X(s)$. Thus,

$$M_Y(s) = e^{\lambda(M_X(s) - 1)} = e^{\lambda(\frac{e^s - 1}{s} - 1)}.$$

- (b) We have using the chain rule

$$\mathbb{E}[Y] = \left. \frac{d}{ds} M_Y(s) \right|_{s=0} = \left. \frac{d}{ds} M_X(s) \right|_{s=0} \cdot \lambda e^{\lambda(M_X(s) - 1)} \Big|_{s=0} = \frac{\lambda}{2},$$

where we have used the fact that $M_X(0) = 1$.

- (c) From the law of iterated expectations we obtain

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|N]] = \mathbb{E}[N\mathbb{E}[X]] = \mathbb{E}[N]\mathbb{E}[X] = \frac{\lambda}{2}.$$

2. Section 4.5 Problem 43.

A motorist goes through 4 lights, each of which is found to be red with probability $1/2$. The waiting times at each light are modeled as independent normal random variables with mean 1 minute and standard deviation $1/2$ minute. Let X be the total waiting time at the red lights.

- (a) Use the total probability theorem to find the PDF and the transform associated with X , and the probability that X exceeds 4 minutes. Is X normal?
- (b) Find the transform associated with X by viewing X as a sum of a random number of random variables.

Solution:

- (a) Using the total probability theorem, we have

$$P(X > 4) = \sum_{k=0}^4 P(k \text{ lights are red})P(X > 4|k \text{ lights are red}).$$

We have

$$P(k \text{ lights are red}) = \binom{4}{k} \left(\frac{1}{2}\right)^4.$$

The conditional PDF of X given that k lights are red, is normal with mean k minutes and standard deviation $(1/2)\sqrt{k}$. Thus, X is a mixture of normal random variables and the transform associated with its (unconditional) PDF is the corresponding mixture of the transforms associated with the (conditional) normal PDFs. However, X is not normal, because a mixture of normal PDFs need not be normal. The probability $P(X > 4|k \text{ lights are red})$ can be computed from the normal tables for each k , and $P(X > 4)$ is obtained by substituting the results in the total probability formula above.

- (b) Let K be the number of traffic lights that are found to be red. We can view X as the sum of K independent normal random variables. Thus the transform associated with X can be found by replacing in the binomial transform $M_K(s) = (1/2 + (1/2)e^s)^4$ the occurrence of e^s by the normal transform corresponding to $\mu = 1$ and $\sigma = 1/2$. Thus

$$M_X(s) = \left(\frac{1}{2} + \frac{1}{2}\left(e^{\frac{(1/2)^2 s^2}{2}} + s\right)\right)^4.$$

Note that by using the formula for the transform, we cannot easily obtain the probability $P(X > 4)$.

3. Section 4.5 Problem 44.

Consider the calculation of the mean and variance of a sum

$$Y = X_1 + \cdots + X_N$$

where N is itself a sum of integer-valued random variables, i.e.,

$$N = K_1 + \cdots + K_M$$

Here $N, M, K_1, K_2, \dots, X_1, X_2, \dots$ are independent random variables, $N, M, K_1, K_2, \dots$ are integer-valued and nonnegative, $K_1, K_2, \dots$ are identically distributed with common mean and variance

denoted $\mathbf{E}[K]$ and $\text{var}(K)$, and $X_1, X_2, \dots$ are identically distributed with common mean and variance denoted $\mathbf{E}[X]$ and $\text{var}(X)$.

- (a) Derive formulas for $\mathbf{E}[N]$ and $\text{var}(N)$ in terms of $\mathbf{E}[M]$, $\text{var}(M)$, $\mathbf{E}[K]$, $\text{var}(K)$.
- (b) Derive formulas for $\mathbf{E}[Y]$ and $\text{var}(Y)$ in terms of $\mathbf{E}[M]$, $\text{var}(M)$, $\mathbf{E}[K]$, $\text{var}(K)$, $\mathbf{E}[X]$, $\text{var}(X)$.
- (c) A crate contains M cartons, where M is geometrically distributed with parameter p . The i th carton contains K_i widgets, where K_i is Poisson-distributed with parameter μ . The weight of each widget is exponentially distributed with parameter λ . All these random variables are independent. Find the expected value and variance of the total weight of a crate.

Solution:

- (a) Using the random sum formulas, we have

$$\begin{aligned}\mathbf{E}[N] &= \mathbf{E}[M]\mathbf{E}[K], \\ \text{var}(N) &= \mathbf{E}[M]\text{var}(K) + (\mathbf{E}[K])^2 \text{var}(M).\end{aligned}$$

- (b) Using the random sum formulas and the results of part (a), we have

$$\begin{aligned}\mathbf{E}[Y] &= \mathbf{E}[N]\mathbf{E}[X] = \mathbf{E}[M]\mathbf{E}[K]\mathbf{E}[X], \\ \text{var}(Y) &= \mathbf{E}[N]\text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N) \\ &= \mathbf{E}[M]\mathbf{E}[K]\text{var}(X) + (\mathbf{E}[X])^2 \left(\mathbf{E}[M]\text{var}(K) + (\mathbf{E}[K])^2 \text{var}(M) \right).\end{aligned}$$

- (c) Let N denote the total number of widgets in the crate, and let X_i denote the weight of the i th widget. The total weight of the crate is

$$Y = X_1 + \dots + X_N,$$

with

$$N = K_1 + \dots + K_M,$$

so the framework of part (b) applies. We have

$$\begin{aligned}\mathbf{E}[M] &= \frac{1}{p}, & \text{var}(M) &= \frac{1-p}{p^2}, & (\text{geometric formulas}), \\ \mathbf{E}[K] &= \mu, & \text{var}(K) &= \mu, & (\text{Poisson formulas}), \\ \mathbf{E}[X] &= \frac{1}{\lambda}, & \text{var}(X) &= \frac{1}{\lambda^2}, & (\text{exponential formulas}).\end{aligned}$$

Using these expressions into the formulas of part (b), we obtain $\mathbf{E}[Y]$ and $\text{var}(Y)$, the mean and variance of the total weight of a crate.

- 4. (a) Let X, Y, Z be independent and uniformly distributed on $[0, 1]$. Find the density function of $X + Y + Z$.

(b) If $\{X_r : r \geq 1\}$ are independent and uniformly distributed on $[0, 1]$, show that the density of $\sum_{r=1}^n X_r$ at any point $x \in (0, n)$ is a polynomial in x of degree $n - 1$.

Solution: The density function of $X + Y$ is, by convolution,

$$f_2(x) = \begin{cases} x, & \text{if } 0 \leq x \leq 1, \\ 2 - x, & \text{if } 1 \leq x \leq 2. \end{cases}$$

Therefore, for $1 \leq x \leq 2$,

$$f_3(x) = \int_0^1 f_2(x - y)dy = \int_{x-1}^1 (x - y)dy + \int_0^{x-1} (2 - x + y)dy = \frac{3}{4} - \left(x - \frac{3}{2}\right)^2.$$

Likewise,

$$f_3(x) = \begin{cases} \frac{1}{2}x^2, & \text{if } 0 \leq x \leq 1, \\ \frac{1}{2}(3 - x)^2, & \text{if } 2 \leq x \leq 3. \end{cases}$$

A simple induction yields the last part.